OPERATIONAL RESEARCH & OPTIMISATION

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Outline

- Introduction to Markov Chain
- Transition probabilities
- Equilibrium

Deterministic optimisation

- A class of optimisation problem where the objective function and constraints are fully known and do not involve randomness:

$$\min_{x \in \mathcal{X}} f(x)$$

- $f : \mathbb{R}^n \to \mathbb{R}$ is a **deterministic** (fixed, non-random) function.
- $\mathcal{X}$ is the feasible set (also fully specified, no randomness).

- But we also face process that involve randomness➔
- Stochastic process

Stochastic process

- A **stochastic process** is a collection of random variables $\{X_t : t \in T\}$ defined on a common probability space, indexed by time (discrete or continuous).
- t is the **index** (can represent time, space, or another dimension).
- X_t is the **random variable** at index t.
- The process describes the **evolution of a system with inherent randomness.**

Stochastic process examples

- Weather
  - Rain, temperature, wind speeds evolve randomly over time.
- Queueing systems
  - Customer arrived at a call center->Poisson process
- Packet arrival in networks
  - Internet traffic can be modelled as stochastic processes
- Stock price evolution
  - Geometric Brownian motion(which is a stochastic process) is the basis of lot models
- ……

Markov Chain

- Is a stochastic process that has a special property (memoryless) and has wide applications, such as in Queueing theory, Finance, Reinforcement learning etc.
- *Acknowledgement: The slides are developed based on Prof. Rachel Fewster's slides for statistics*

It only matters where you are, not where you've been…

A.A.Markov
1856-1922

Stochastic process definition-1

Definition: A **stochastic process** is a *family of random variables*, $\{X(t) : t \in T\}$, *where t usually denotes time. That is, at every time t in the set T, a random number $X(t)$ is observed.*

Definition: $\{X(t) : t \in T\}$ is a **discrete-time process** if *the set T is finite or countable.*

In practice, this generally means $T = \{0, 1, 2, 3, \ldots\}$

Thus a discrete-time process is $\{X(0), X(1), X(2), X(3), \ldots\}$*: a random number associated with every time 0, 1, 2, 3, ...*

Stochastic process definition-2

Definition: The **state space**, S, is ***the set of real values that $X(t)$ can take.***

Every $X(t)$ takes a value in $\mathbb{R}$, but S will often be a smaller set: $S \subseteq \mathbb{R}$. For example, if $X(t)$ is the outcome of a coin tossed at time t, then the state space is $S = \{0, 1\}$.

Markov Chain definition-1

The Markov chain is the process $X_0, X_1, X_2, \ldots$.

Definition: The **state** of a Markov chain at time t is the ***value of*** X_t.

For example, if $X_t = 6$, we say ***the process is in state*** 6 ***at time*** t.

Definition: The **state space** of a Markov chain, S, is the set of values that each X_t can take. For example, $S = \{1, 2, 3, 4, 5, 6, 7\}$.

Let S have size N (possibly infinite).

Markov Chain definition-2

Definition: A **trajectory** of a Markov chain is ***a particular set of values for*** $X_0, X_1, X_2, \ldots$.

For example, if $X_0 = 1$, $X_1 = 5$, and $X_2 = 6$, then the trajectory up to time $t = 2$ is $1, 5, 6$.

More generally, if we refer to the trajectory $s_0, s_1, s_2, s_3, \ldots$, we mean that $X_0 = s_0, X_1 = s_1, X_2 = s_2, X_3 = s_3, \ldots$

'Trajectory' is just a word meaning ***'path'***.

Markov Chain- A simple example

- Weather model:
- We use X_i, i=0, 1, 2, 3.. to denote the weather for day 0, 1, 2, 3...
- Suppose weather can be either sunny or rainy, thus the state space={Hot, Cold}
- A trajectory of the weather Markov chain could be X_0 =Hot, X_1=Hot, X_2=Cold…

Markov Property-1

The basic property of a Markov chain is that *only the most recent point in the trajectory affects what happens next.*

This is called the *Markov Property.*
It means that X_{t+1} *depends upon* X_t, *but it does not depend upon* $X_{t-1}, \ldots, X_1, X_0$.

Markov Property-2

We formulate the Markov Property in mathematical notation as follows:

$$\boxed{\mathbb{P}(X_{t+1} = s \mid X_t = s_t, X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) \;=\; \mathbb{P}(X_{t+1} = s \mid X_t = s_t),}$$

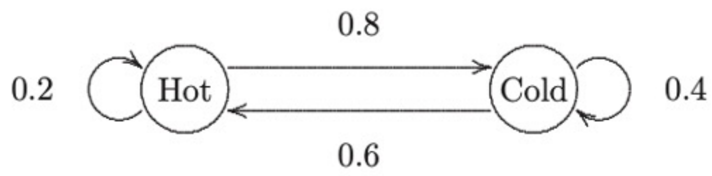
for all $t = 1, 2, 3, \ldots$ and for all states $s_0, s_1, \ldots, s_t, s$.

Explanation:

$$\underset{\substack{\uparrow \\ \textit{distribution} \\ \textit{of } X_{t+1}}}{\mathbb{P}(X_{t+1} = s} \quad \Big| \quad \underset{\substack{\uparrow \\ \textit{depends} \\ \textit{on } X_t}}{X_t = s_t,} \quad \underbrace{\cancel{X_{t-1} = s_{t-1}, X_{t-2} = s_{t-2}, \ldots, X_1 = s_1, X_0 = s_0}}_{\substack{\uparrow \\ \textit{but whatever happened before time } t \\ \textit{doesn't matter.}}})$$

Transition probabilities

- From above formulation, a Markov chain *cannot be in two states at the same time*.
- It can however change states from one state to another. When this happens, it is called a *transition* from state s_n to state s_{n+1}.
- This transition is represented with *a probability value*.
- See below the weather example:



State transition matrix

- The *state transition matrix* is a matrix of all transition probabilities.

Transition Matrix

$$\begin{array}{cc} & \overbrace{\longleftarrow \text{list all states} \longrightarrow}^{X_{t+1}} \\ X_t \left\{ \begin{array}{c}\text{list}\\ \text{all}\\ \text{states}\end{array}\right. \updownarrow & \left(\begin{array}{c} \texttt{insert} \\ \texttt{probabilities} \\ p_{ij} \end{array} \right) \begin{array}{l} \longleftarrow \text{rows add to 1} \\ \longleftarrow \text{rows add to 1}\end{array} \end{array}$$

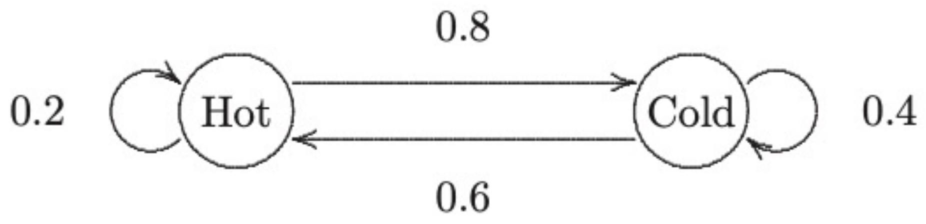
The transition matrix is usually given the symbol $P = (p_{ij})$.

In the transition matrix P:

- *the ROWS represent NOW, or FROM (X_t);*
- *the COLUMNS represent NEXT, or TO (X_{t+1});*
- *entry (i, j) is the CONDITIONAL probability that NEXT $= j$, given that NOW $= i$: the probability of going FROM state i TO state j.*

$$\boxed{p_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i).}$$

Transition matrix:



summarize the probabilities

$$
X_t \left\{ \begin{array}{c} \\ \text{Hot} \\ \text{Cold} \end{array} \right.
\begin{array}{c}
\overbrace{\begin{array}{cc} \text{Hot} & \text{Cold} \end{array}}^{X_{t+1}} \\
\begin{pmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{pmatrix}
\end{array}
$$

Transition matrix

Notes: 1. The transition matrix P must list *all* possible states in the state space S.

2. P is a *square matrix* $(N \times N)$, because X_{t+1} and X_t both take values in the same state space S (of size N).

3. The **rows** of P should each ***sum to 1:***

$$\sum_{j=1}^{N} p_{ij} = \sum_{j=1}^{N} \mathbb{P}(X_{t+1} = j \mid X_t = i) = \sum_{j=1}^{N} \mathbb{P}_{\{X_t = i\}}(X_{t+1} = j) = 1.$$

This simply states that X_{t+1} *must* take one of the listed values.

4. The **columns** of P do **not** in general sum to 1.

Another transition matrix example

- Consider a simple economy prediction model where states correspond to *inflation* or *deflation* in given economic cycles.
- The states represent whether the hypothetical economy is currently undergoing inflation or deflation in a specific season.
- In our model, inflation is followed by either another inflation with 95% probability or deflation with 5% probability. Deflation is followed by either inflation with 80% probability or another deflation with 20% probability.

Economic model transition matrix

- Try to write down the transition matrix:

The t-step transition probabilities-1

Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with state space $S = \{1, 2, \ldots, N\}$.

Recall that the elements of the transition matrix P are defined as:

$$(P)_{ij} = p_{ij} = \mathbb{P}(X_1 = j \,|\, X_0 = i) = \mathbb{P}(X_{n+1} = j \,|\, X_n = i) \quad \text{for any } n.$$

p_{ij} is the probability of making a transition FROM state i TO state j in a SINGLE step.

Question: what is the probability of making a transition from state i to state j over **two** steps? *I.e. what is* $\mathbb{P}(X_2 = j \,|\, X_0 = i)$*?*

The t-step transition probabilities-2

Theorem : Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with $N \times N$ transition matrix P. Then the t-step transition probabilities are given by the matrix P^t. That is,

$$\mathbb{P}(X_t = j \mid X_0 = i) = \left(P^t\right)_{ij}.$$

It also follows that

$$\mathbb{P}(X_{n+t} = j \mid X_n = i) = \left(P^t\right)_{ij} \text{ for any } n. \qquad \square$$

- *where P^T denotes that matrix P multiply itself T times*

Economic model transition matrix example

- One step transition matrix:
- $P^1 = \begin{pmatrix} 0.95 & 0.05 \\ 0.80 & 0.20 \end{pmatrix}$
- Two-step transition matrix:
- $P^2 = \begin{pmatrix} 0.95 & 0.05 \\ 0.80 & 0.20 \end{pmatrix} \begin{pmatrix} 0.95 & 0.05 \\ 0.80 & 0.20 \end{pmatrix}$

$$= \begin{pmatrix} 0.95 * 0.95 + 0.05 * 0.80 & 0.95 * 0.05 + 0.05 * 0.20 \\ 0.80 * 0.95 + 0.20 * 0.80 & 0.80 * 0.05 + 0.20 * 0.20 \end{pmatrix}$$

$$= \begin{pmatrix} 0.9425 & 0.0575 \\ 0.92 & 0.08 \end{pmatrix}$$

You can do the same to get $P^3, P^4 \ldots$

Review matrix multiplication from your level 4 math module

Probability distribution of $X_0, X_1, X_2, \ldots$

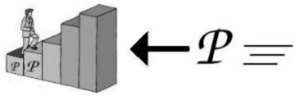
<u>**Theorem :**</u> Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with $N \times N$ transition matrix P. If the probability distribution of X_0 is given by the $1 \times N$ row vector $\boldsymbol{\pi}^T$, then the probability distribution of X_t is given by the $1 \times N$ row vector $\boldsymbol{\pi}^T P^t$. That is,

$$X_0 \sim \boldsymbol{\pi}^T \quad \Rightarrow \quad X_t \sim \boldsymbol{\pi}^T P^t.$$

Note: The distribution of X_t is $X_t \sim \boldsymbol{\pi}^T P^t$.
The distribution of X_{t+1} is $X_{t+1} \sim \boldsymbol{\pi}^T P^{t+1}$.
Taking one step in the Markov chain corresponds to ***multiplying by*** P ***on the right.***

Note: The t-step transition matrix is P^t .
The $(t+1)$-step transition matrix is P^{t+1}.
Again, taking one step in the Markov chain corresponds to ***multiplying by*** P ***on the right.***

take 1 step...



...multiply by P on the right

Economic model example

- If the initial state $X_0 = (0.90, 0.10)$, then what will be the economic state after 2 economic cycles(2 time steps)?
- Solution:
- $X_2 = X_0 \begin{pmatrix} 0.95 & 0.05 \\ 0.80 & 0.20 \end{pmatrix}^2$
- $= (0.90{,}0.10) \begin{pmatrix} 0.9425 & 0.05 \\ 0.8 & 0.0425 \end{pmatrix}$
- $= (0.935{,}0.065)$

Equilibrium: steady-state

<u>**Theorem :**</u> Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with $N \times N$ transition matrix P. If the probability distribution of X_0 is given by the $1 \times N$ row vector $\boldsymbol{\pi}^T$, then the probability distribution of X_t is given by the $1 \times N$ row vector $\boldsymbol{\pi}^T P^t$. That is,

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Note: The distribution of X_t is $X_t \sim \boldsymbol{\pi}^T P^t$.
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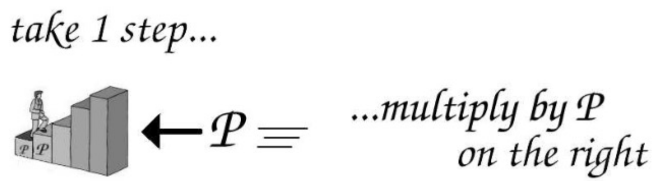
Taking one step in the Markov chain corresponds to ***multiplying by P on the right***.

Note: The t-step transition matrix is P^t .
The $(t+1)$-step transition matrix is P^{t+1}.
Again, taking one step in the Markov chain corresponds to ***multiplying by P on the right***.



Equilibrium: steady-state

This raises the question: is there any distribution $\boldsymbol{\pi}$ such that $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$?

If $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$, then

$$\begin{aligned} X_t \sim \boldsymbol{\pi}^T \quad & \Rightarrow \quad X_{t+1} \sim \boldsymbol{\pi}^T P = \boldsymbol{\pi}^T \\ & \Rightarrow \quad X_{t+2} \sim \boldsymbol{\pi}^T P = \boldsymbol{\pi}^T \\ & \Rightarrow \quad X_{t+3} \sim \boldsymbol{\pi}^T P = \boldsymbol{\pi}^T \\ & \Rightarrow \quad \ldots \end{aligned}$$

In other words, if $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$, and $X_t \sim \boldsymbol{\pi}^T$, then

$$X_t \sim X_{t+1} \sim X_{t+2} \sim X_{t+3} \sim \ldots$$

Thus, once a Markov chain has reached a distribution $\boldsymbol{\pi}^T$ such that $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$, ***it will stay there.***

If $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$, we say that the distribution $\boldsymbol{\pi}^T$ is an ***equilibrium distribution.***

Equilibrium means a ***level position:*** there is *no more change* in the distribution of X_t as we wander through the Markov chain.

Equilibrium: steady-state

Note: Equilibrium does not mean that the **value** of X_{t+1} equals the **value** of X_t.
It means that the **distribution** of X_{t+1} is the same as the **distribution** of X_t:

e.g. $\mathbb{P}(X_{t+1} = 1) = \mathbb{P}(X_t = 1) = \pi_1$;

$\mathbb{P}(X_{t+1} = 2) = \mathbb{P}(X_t = 2) = \pi_2$, *etc.*

- *How to calculate the equilibrium distribution?*
- Many Markov chains automatically find their own way to an equilibrium distribution as the chain wanders through time. This happens for many Markov chains, but not all.

Calculating equilibrium(steady-state) distributions π_i

Definition: Let $\{X_0, X_1, \ldots\}$ be a Markov chain with transition matrix P and state space S, where $|S| = N$ (possibly infinite). Let $\boldsymbol{\pi}^T$ be a row vector denoting a probability distribution on S: so each element π_i denotes the probability of being in state i, and $\sum_{i=1}^{N} \pi_i = 1$, where $\pi_i \geq 0$ for all $i = 1, \ldots, N$. The probability distribution $\boldsymbol{\pi}^T$ is an **equilibrium** distribution for the Markov chain if $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$.

That is, $\boldsymbol{\pi}^T$ is an equilibrium distribution if

$$\left(\boldsymbol{\pi}^T P\right)_j = \sum_{i=1}^{N} \pi_i p_{ij} = \pi_j \quad \textit{for all } j = 1, \ldots, N.$$

Calculating equilibrium(steady-state) distributions π_i

Vector $\boldsymbol{\pi}^T$ is an equilibrium distribution for P if:

1. $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$;
2. $\sum_{i=1}^{N} \pi_i = 1$;
3. $\pi_i \geq 0$ *for all* i.

Calculating equilibrium(steady-state) distributions π_i

- Economic state example:

We know that 1 step transition matrix is

$$\begin{pmatrix} 0.95 & 0.05 \\ 0.80 & 0.20 \end{pmatrix}$$

Let π^T be the equilibrium distribution vector.

$\pi^T = (\pi_0, \pi_1)$ where π_0, π_1 are scaler.

then follow previous slides condition we have

Calculating equilibrium(steady-state) distributions π_i

- Economic state example:

$$(\pi_0, \pi_1)\begin{pmatrix} 0.95 & 0.05 \\ 0.80 & 0.20 \end{pmatrix} = (\pi_0, \pi_1) \qquad ①.$$

$$\pi_0 + \pi_1 = 1 \qquad ②$$

$$\pi_0, \pi_1 \geq 0 \qquad ③$$

We perform the matrix multiplication on left hand side

We have

Calculating equilibrium(steady-state) distributions π_i

- Economic state example:

$$(0.95\pi_0 + 0.80\pi_1,\ 0.05\pi_0 + 0.20\pi_1) = (\pi_0, \pi_1) \quad ④$$

from ④ we have

$$0.95\pi_0 + 0.80\pi_1 = \pi_0 \quad ⑤$$

$$0.05\pi_0 + 0.20\pi_1 = \pi_1 \quad ⑥$$

together with ②, ③

we have

Calculating equilibrium(steady-state) distributions π_i

- Economic state example:

$$\begin{cases} 0.95\pi_0 + 0.80\pi_1 = \pi_0 \\ 0.05\pi_0 + 0.20\pi_1 = \pi_1 \\ \pi_0 + \pi_1 = 1 \end{cases}$$

Solve this system of equations simultaneously we have

$$0.95\pi_0 + 0.80(1 - \pi_0) = \pi_0$$

thus $0.95\pi_0 + 0.80 - 0.80\pi_0 = \pi_0$

$$0.95\pi_0 = 0.80$$

$$\pi_0 = \frac{80}{95} = \frac{16}{17}$$

Calculating equilibrium(steady-state) distributions π_i

- Economic state example:

thus $\pi_1 = 1 - \pi_0 = 1 - \frac{16}{17} = \frac{1}{17}$

Calculating equilibrium(steady-state) distributions π_i

- Each row of the transition matrix P^t converges to the equilibrium distribution (16/17,1/17) or (0.941,0.059) as $t \to \infty$
- This convergence of P^t means that for large t, no matter **which** state we start in, we always have probability 16/17 of being inflation after t steps, 1/17 of being deflation after t steps.

Extension

- *Queueing theory* applies Markov chain models to analyse systems with random arrivals and service times.
- The *Markov property* makes it mathematically tractable to derive *performance metrics (e.g. Arrival rate, Service rate, The average number of customers in the system etc.)* and *steady-state behaviour*.
- Next week: how to *optimise the performance* of such system.

- Q&A